



RUHR-UNIVERSITÄT BOCHUM

INTRODUCTION

Computer Architecture

Agenda

- Organization
- What is this course about?
 - Background
 - Course outlook
- Overarching topics
 - Moore's law
 - Amdahl's Law

Teacher

Yuval Yarom

Office: MC 5/148

email: yuval.yarom@rub.de

Consulting: Monday 14:00-15:30

No face recognition



The Team



Mariia Charkina



Luis Kosel



Ole Volkening



Birk Wildhirt



Sarah Corneli



Lenja Limberg



Marius Kleinert



Natalie Riemer

Moodle

Moodle Course: <https://moodle.ruhr-uni-bochum.de/course/view.php?id=65690>

- Course information
- Resources
- Lecture slides
- Lecture recordings
- Exercise sheets
- Weekly assignments
- Announcements
- Q&A forum
- Contact Details



Course

Lecture: Friday 10:15-11:45 HZO 30

Tutorials:

Tuesday	08:15-09:45 ID 03/463; MC 1/30+31
Tuesday	10:15-11:45 MB 0/105
Tuesday	12:15-13:45 ID 03/411
Wednesday	10:15-11:45 ID 03/463
Wednesday	12:15-13:45 ID 03/471; IA 01/473
Thursday	10:15-11:45 MB 0/105
Friday	14:15-15:45 ID 03/463; MC 1/30+31



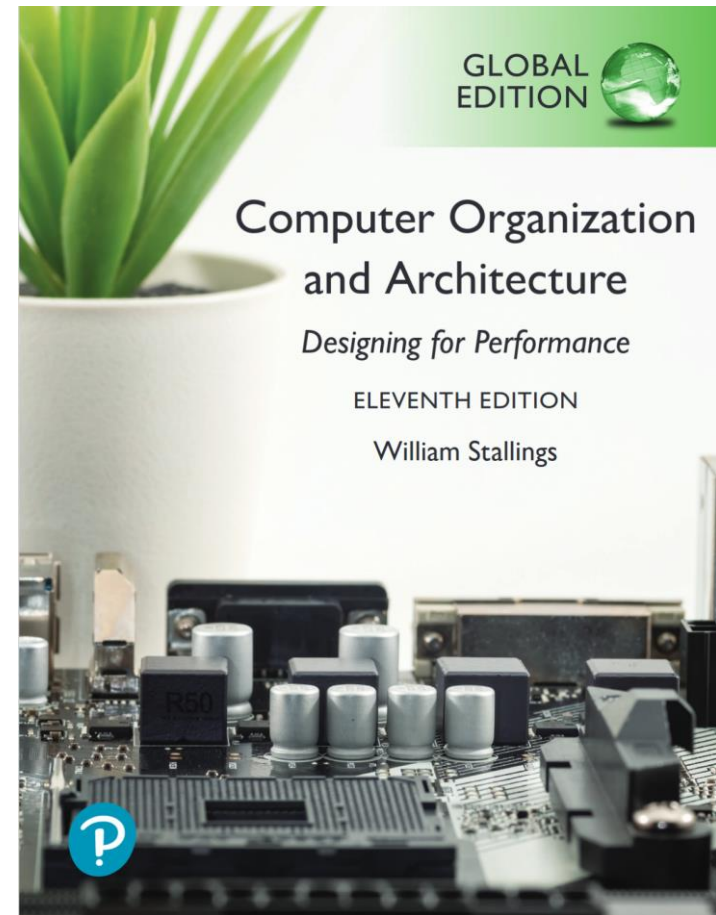
You pick one

Help Desk: Wednesday 14:00-15:00, MC 5/144

Textbook

William Stallings, Computer Organization and Architecture

- Four copies in the library



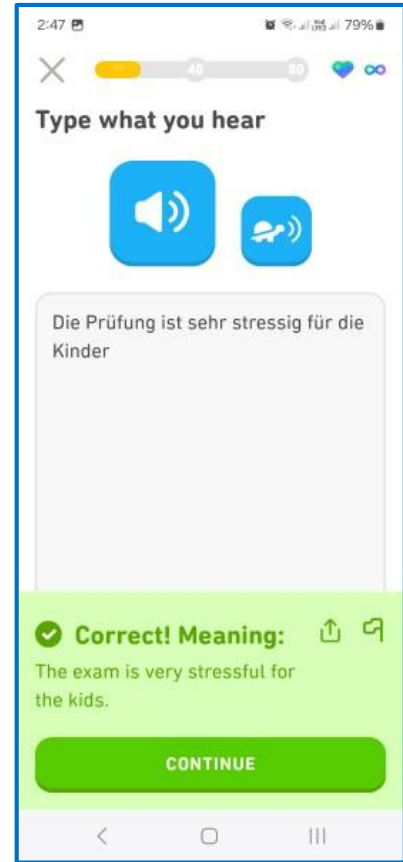
Assessment

Exam

- Written exam, 2 hours.
- Follow the examination office rules to register/deregister.
- If sick after deregistration deadline, contact the examination office.
 - See: <https://informatik.rub.de/en/studium/examination-office/illness-during-exams/>

Weekly Assignments

- Bonus is the average of the assignment grades mapped to the range 0-10, at a 0.5 point interval
- Maximum final grade 100%
- Bonus does not apply if you fail the exam
- Bonus will be revoked if you are caught cheating



Learning Support

- Lecture recordings
 - Best effort – had technical issues last year
- Tutorials
- Assignments
 - Bonus points are the smaller benefit
- Practice questions and exams
- Q&A forum
- Help desk
- Consultation hours

GradeScope

- We use GradeScope to mark assignments and exams
 - Q&A-Tasks: Use provided template and upload PDF to GradeScope
 - Programming tasks: upload code as .zip to GradeScope
 - Online marking and reviews
- Information shared with GradeScope
 - Email addresses (which implies names)
 - Matriculation number (exam only)
 - Assignment and exam contents and marks
- European servers
- The University's Data Protection Officer approves the Data Processing Agreement
- Gradescope's Privacy Policy

Academic Integrity

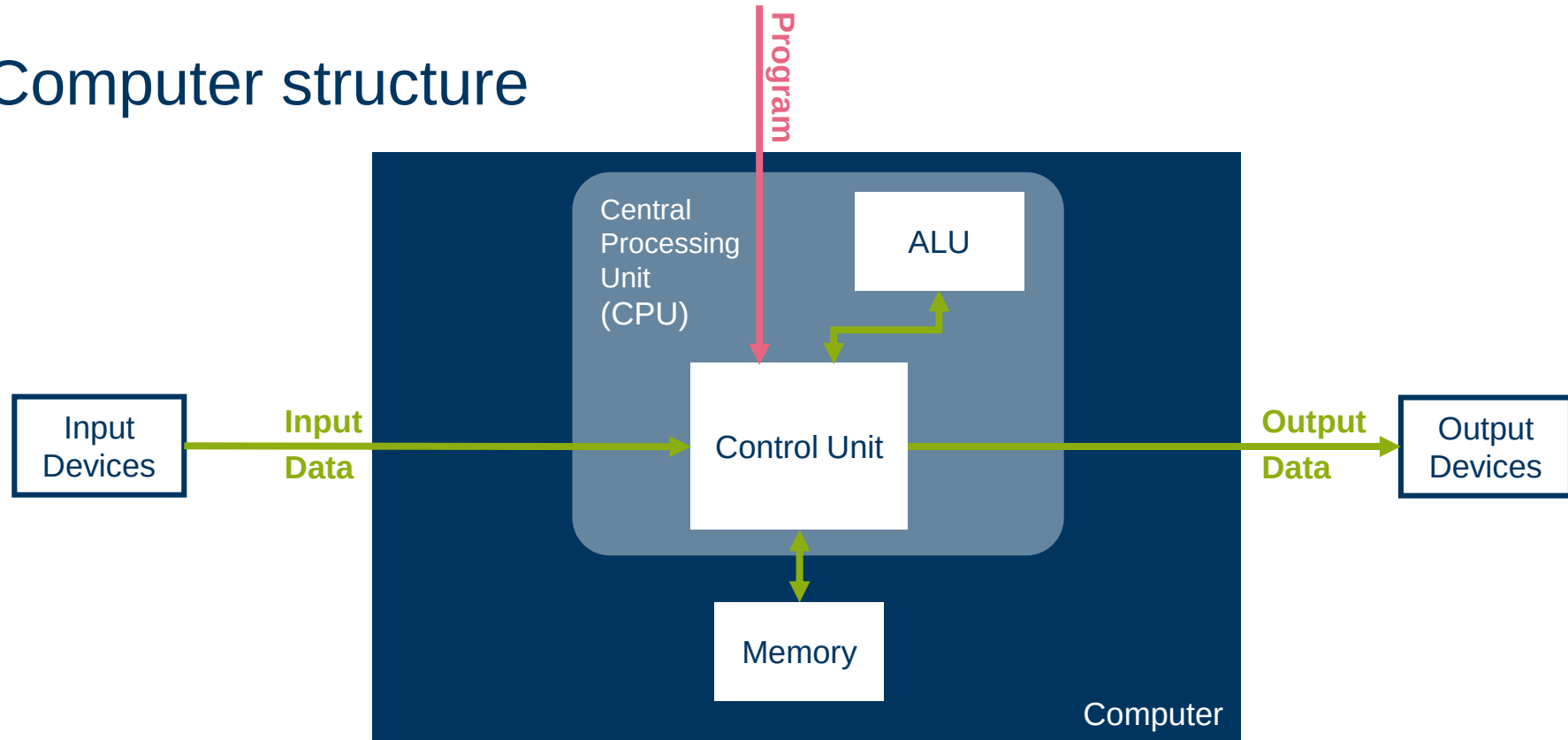
- Assignments are individual
 - Do not work with friends
 - Do not copy from other students
 - Do not allow other students to copy your work
- Avoid plagiarism
 - Cite sources when copying or paraphrasing
- **Don't use AI**



Image generated with AI using OpenAI's DALL·E. © 2025 OpenAI. All rights reserved.

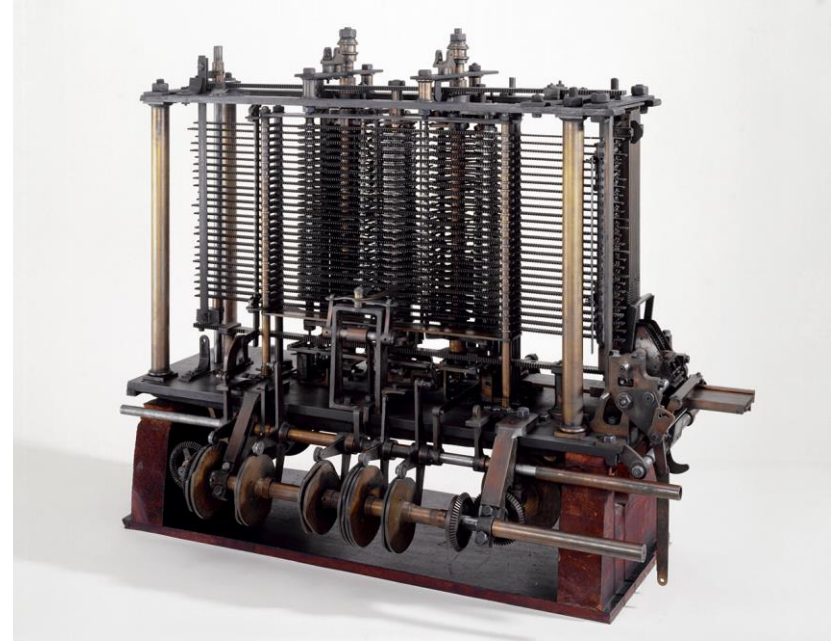
Introduction

Computer structure



The Analytical Engine (1837)

- Mechanical computer
- Designed by Charles Babbage
- Never completed



Images: Charles Babbage/ Karoly Lorentey
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Harvard Mark 1 (1944)

Electromechanical general-purpose computer

Designed by IBM and Harvard

Used for calculations necessary for the Manhattan Project

Could store up to 72 numbers of 23 decimal digits each at the same time

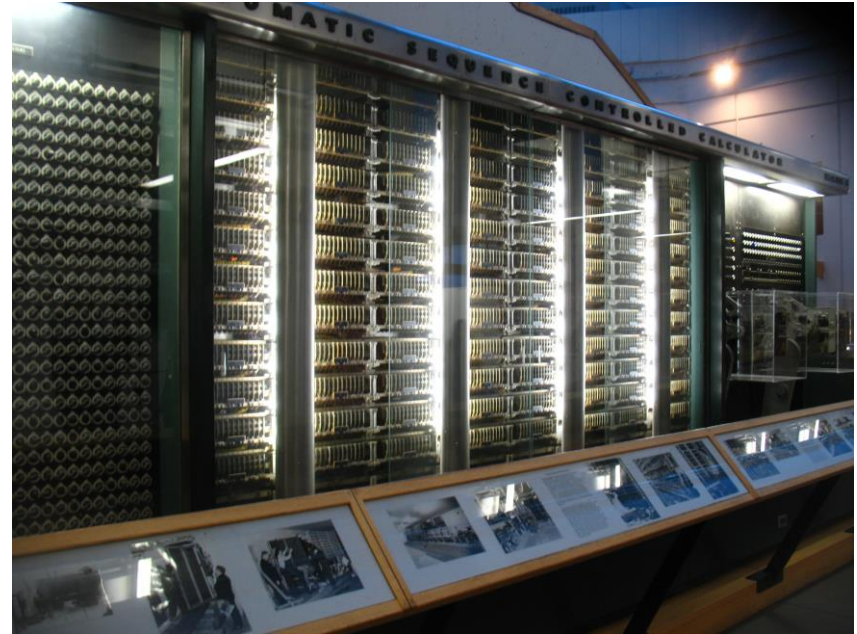
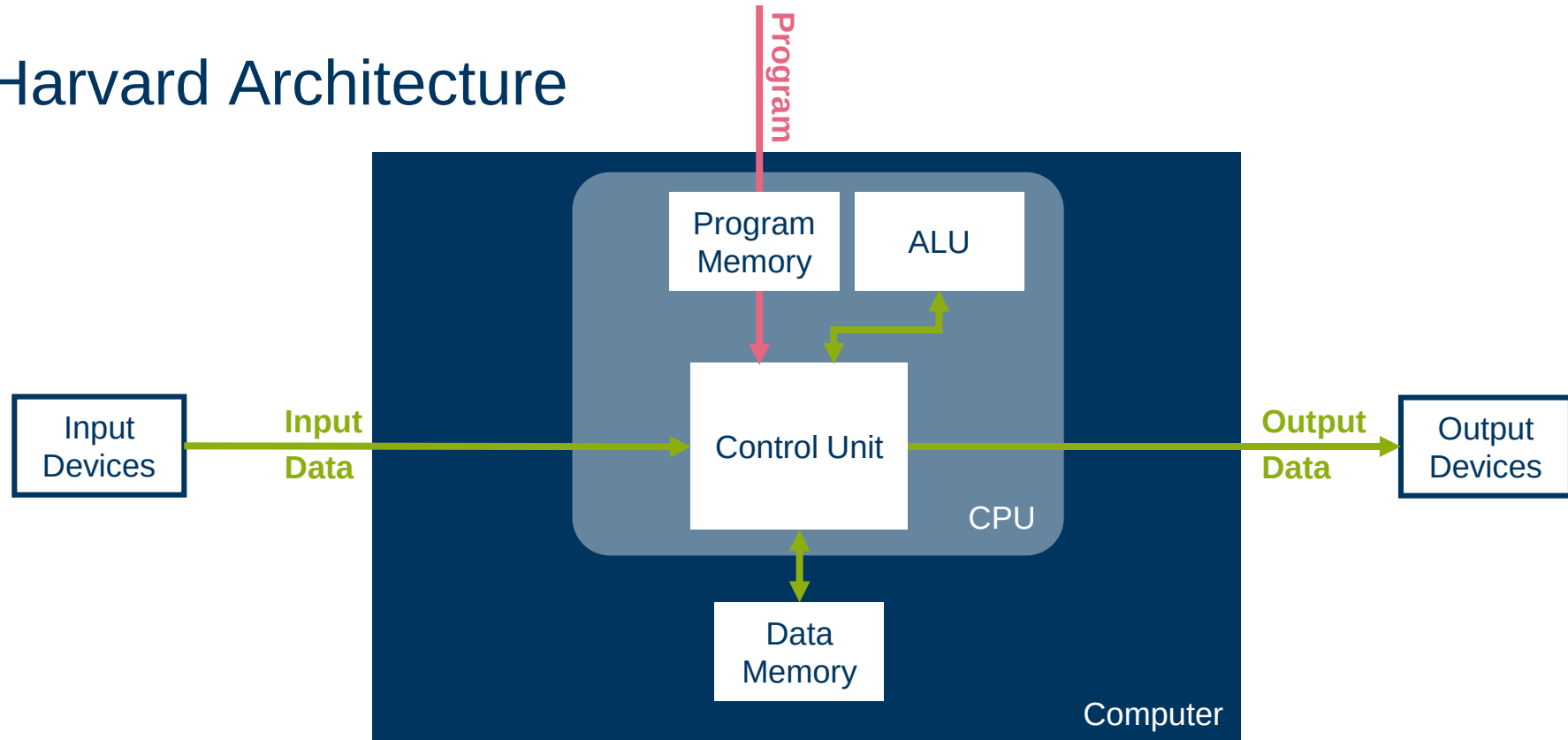
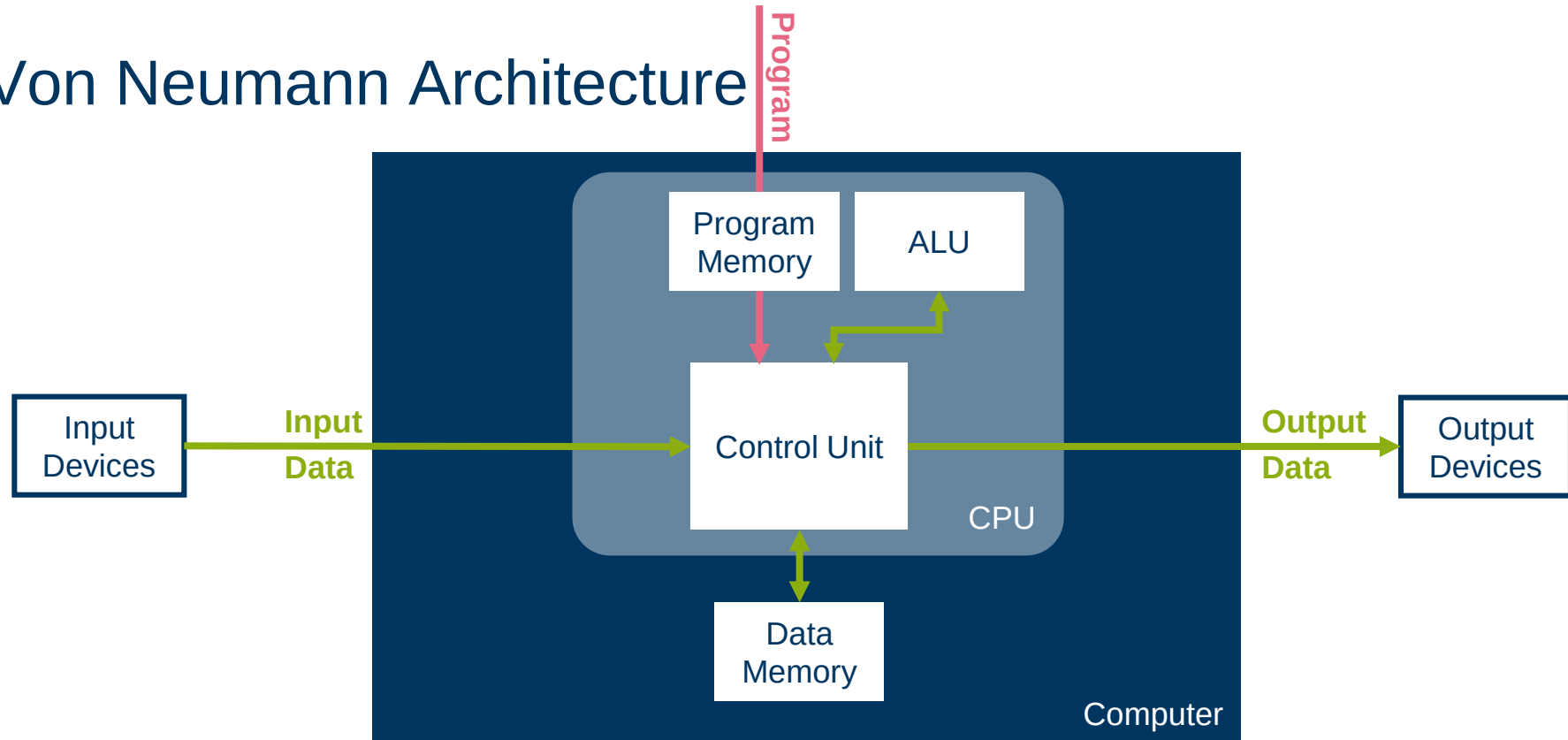


Photo by Sara D. / CC BY 2.0

Harvard Architecture

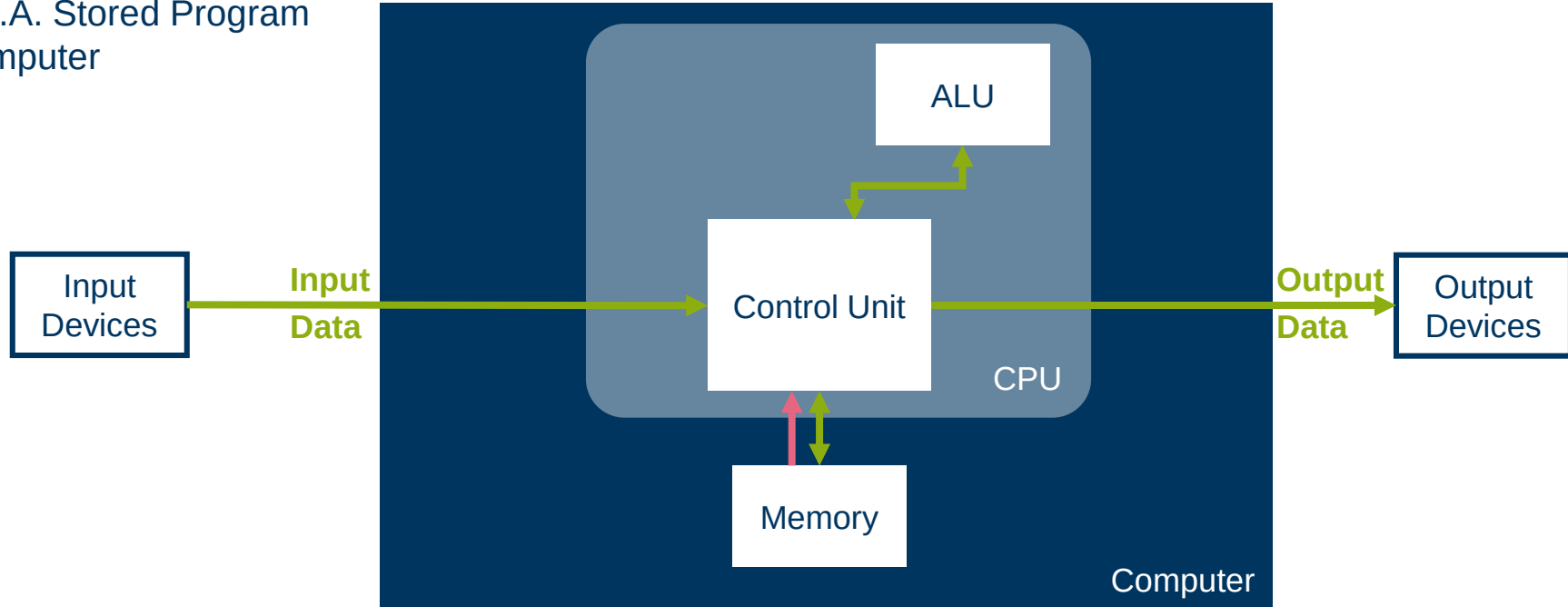


Von Neumann Architecture



Von Neumann Architecture

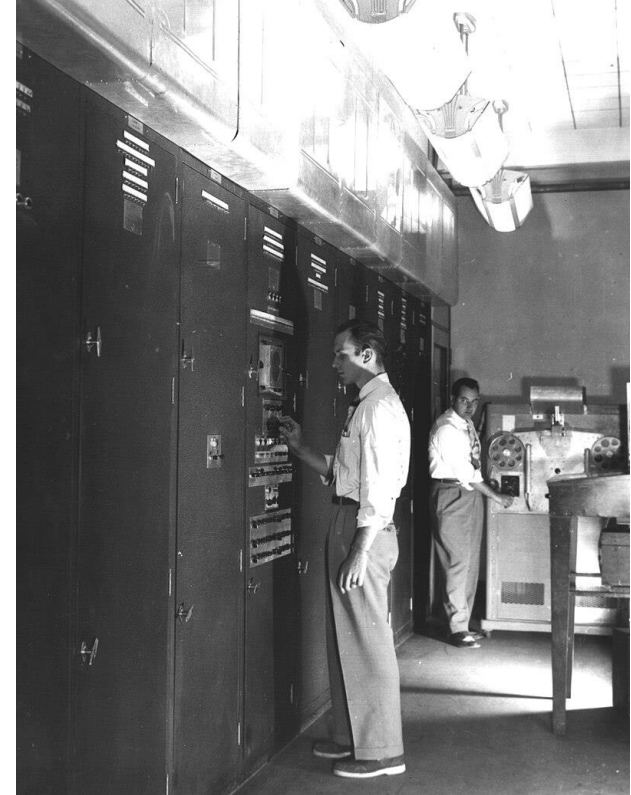
A.K.A. Stored Program
Computer



EDVAC (1949)

First stored program computer

- Ultrasonic serial memory:
 - Mercury acoustic delay lines
 - 1024 word of 44 bits each
- 864 μ s for addition, 2900 μ s multiplication



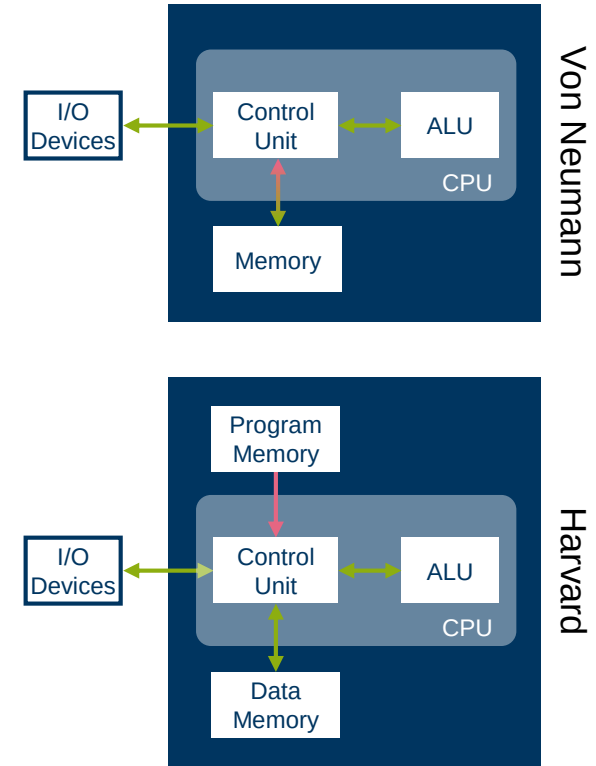
Harvard vs. von Neumann

“Program is data” offers flexibility

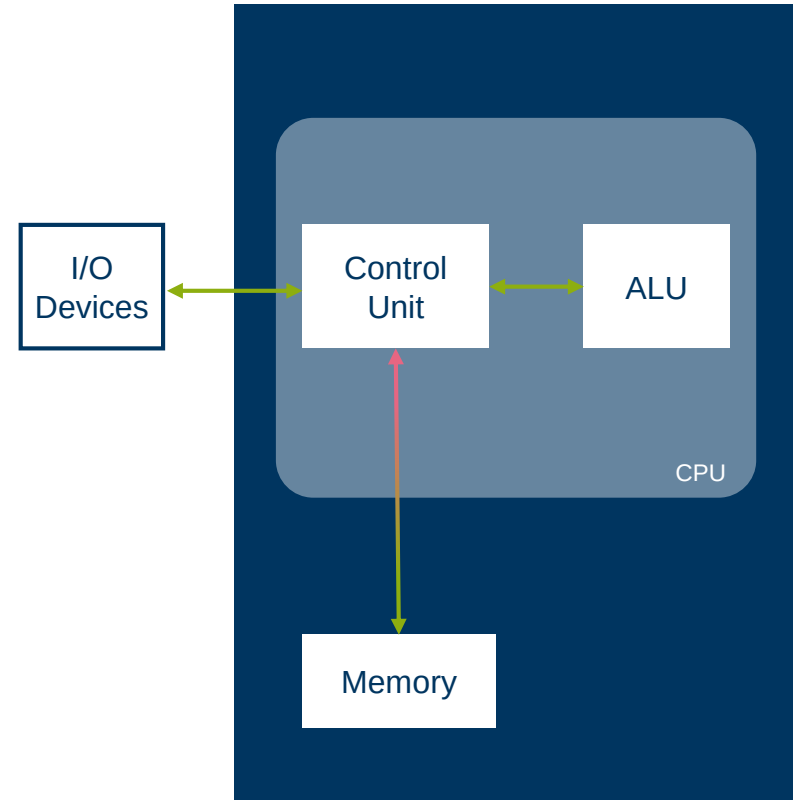
- Loading programs from I/O devices
- Generating code on the fly
- Self-modifying code

Single bus is

- Simpler
- More space and power efficient
- Bottleneck



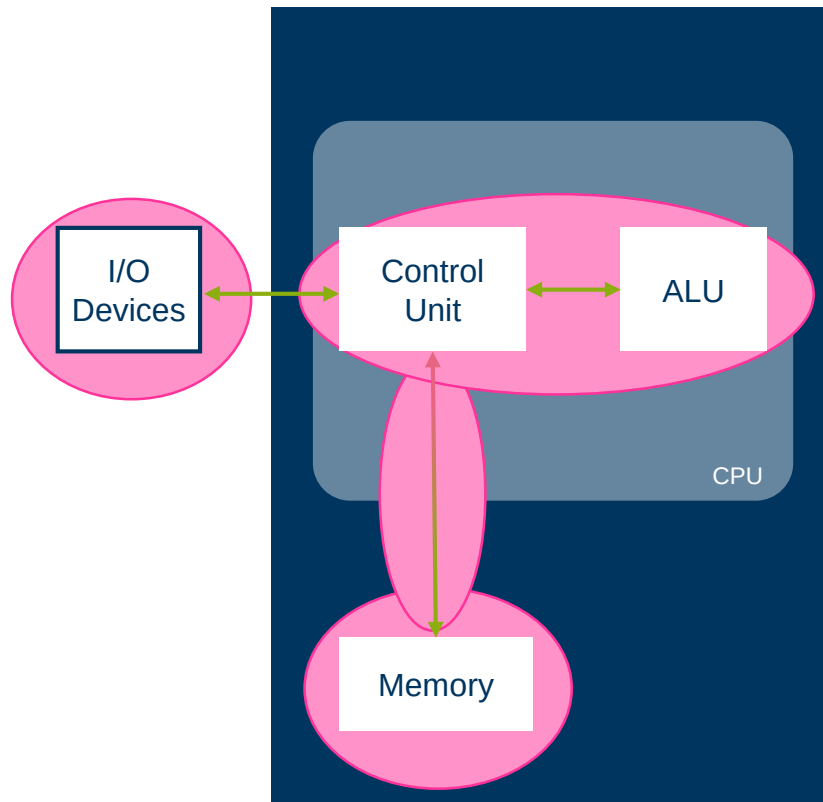
Modern Computers



Course Outlook

- Introduction – 1 lecture
- Data/code representation – 3 lectures
- The memory subsystem – 2 lectures
- Execution pipeline – 3 lectures
- I/O devices – 2 lectures
- Security – 1 lecture
- Exam revision – 1 lecture

Changes are possible!

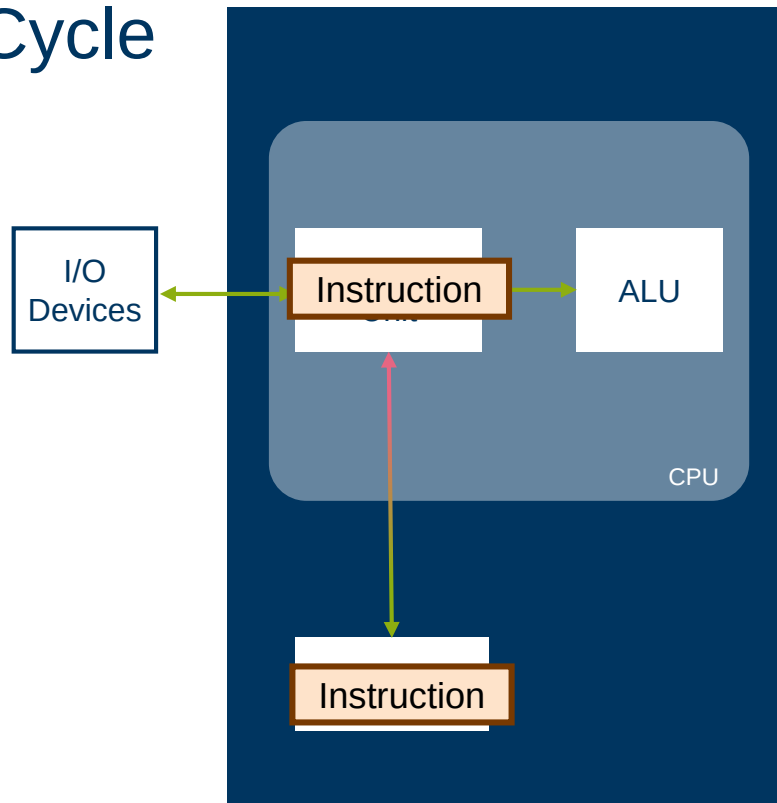


Operating principles

The Fetch-Decode-Execute Cycle

Program execution follows three main steps:

- **Fetch:** bring the next instruction from memory to control unit
- **Decode:** “understand” what the instruction needs to do and set up for execution
- **Execute:** perform the instruction

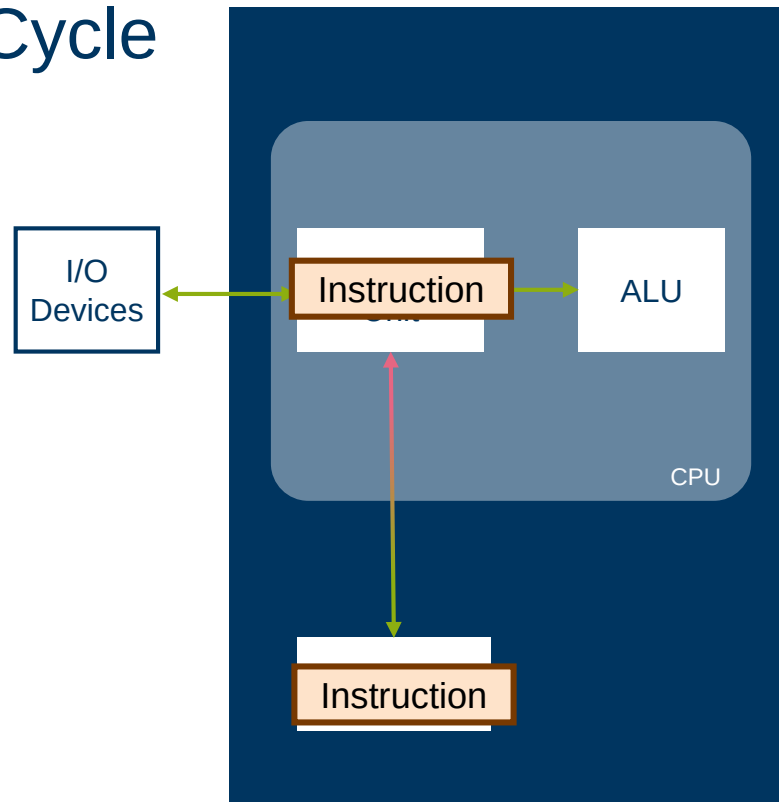


The Fetch-Decode-Execute Cycle

Program execution follows three main steps:

- **Fetch:** bring the next instruction from memory to control unit
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- **Execute:** perform the instruction

Repeat



Example

I've just arrived in Bochum
and would like to visit the
Bergbaumuseum

I don't have a map and my
phone is dead

I do have some instructions



Tuxyso / Wikimedia Commons / CC BY-SA 3.0

Example

Geh über die Straße und dann nach rechts

Biege an der Kreuzung nach links ab

Am Hotel musst du nach rechts gehen

Jetzt musst du bis zur Schule geradeaus gehen und dann nach links abbiegen

Jetzt gehst du 900 Meter geradeaus und bist da

decode

Cross the street and then turn right

Go left at the crossroads

When you reach the hotel, go right

Now you need to walk straight on.
When you reach the school, turn left

Walk straight on for about 900 meters and you should be there

Example

I have arrived at the
Bergbaumuseum!



Tuxyso / Wikimedia Commons / CC BY-SA 3.0

Moore's Law

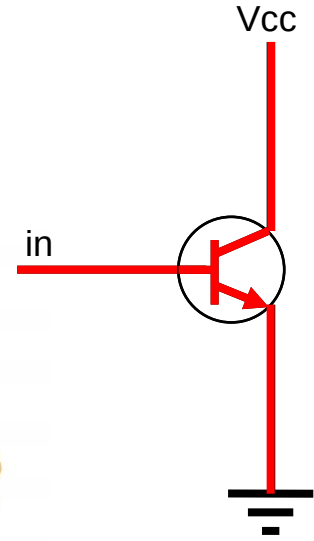
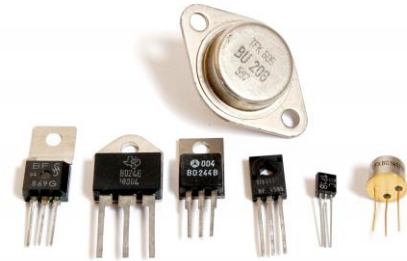
The Transistors

Full electric switch

Voltage at Gate decides if a current can flow

Used in logic gates, memory storage

Crucial component for today's computers



Moore's Law (1965)

The number of transistors in an integrated circuit is going to double every year

- Gordon Moore, co-founder of Intel



Photo by OnInnovation/ CC BY-ND 2.0

Moore's Law Over the Years

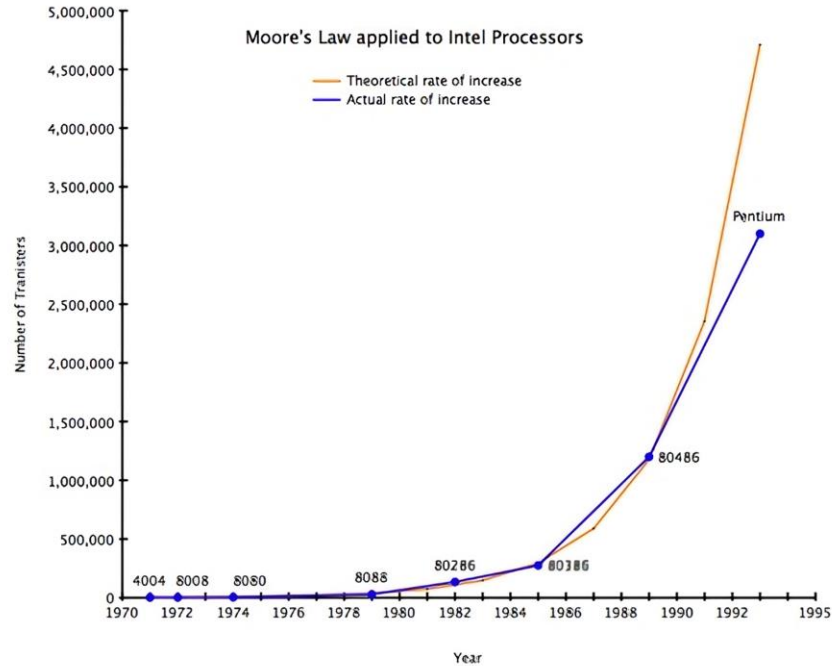


Photo by Bilby/ CC BY-SA 3.0

Our World
in Data

The End Of Moore's Law

1975 – Correction: The number of transistors is going to double every two years

2015 – Intel: One cycle takes up to 30 months now

Moore's Law is predicted to end around 2025

More Than Moore?

Transistor minimization approaches its physical limitations

Shift focus from transistor density to integration of additional components:

- sensors
- communication modules
- GPUs
- [...]

Parallelization

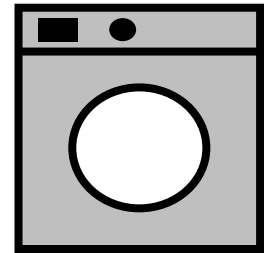
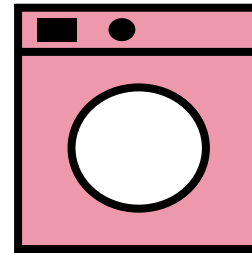
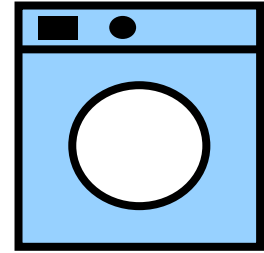
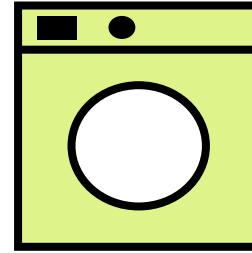
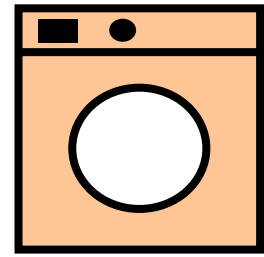
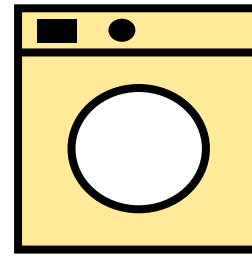
Doing Laundry

A load of laundry takes one hour.

We also need 5 minutes to load and 5 to unload

How long does it take to do 6 loads?

How much faster will we be if we have six washing machines?



Amdahl's Law

We have a job that we want to complete.

We want to add resources, so it completes faster.

How much faster can we expect the job to be?
What is the expected speedup?

The performance improvement of parallelization is limited by the percentage of tasks that have to be performed sequentially
-Gene Amdahl



Photo by Pkivowitz/ CC BY 3.0

Amdahl's Law

Without additional resources our job takes time T to complete

Parallelizable tasks are $p\%$ of the job

With added resources we can speed the parallelizable tasks by a factor of s

Original time for parallelizable tasks: pT

Original time for non-parallelizable tasks: $(1 - p)T$

New time for parallelizable tasks: $\frac{p}{s}T$

New time for job: $(1 - p)T + \frac{p}{s}T$

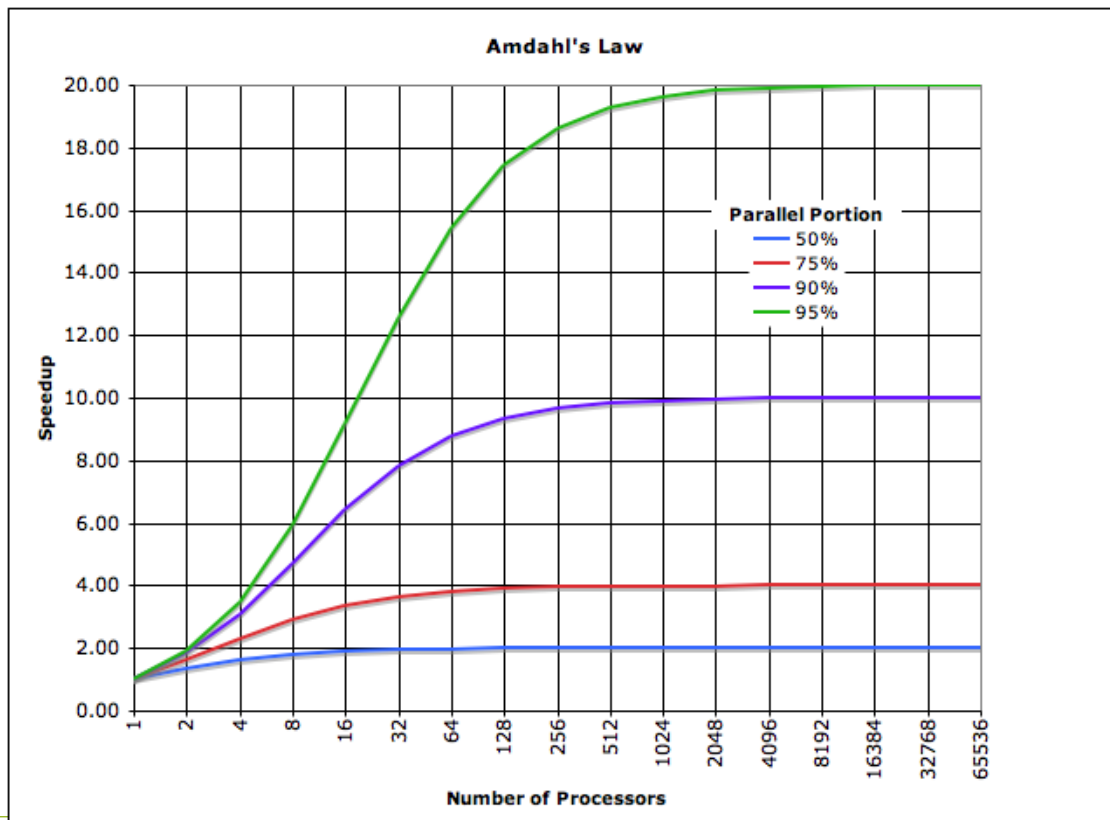
Amdahl's Law

New time for job: $(1 - p)T + \frac{p}{s}T$

We can now compute the speedup S

$$S = \frac{T}{(1-p)T + \frac{p}{s}T} = \frac{T}{T(1-p + \frac{p}{s})}$$

$$S = \frac{1}{1 - p + \frac{p}{s}}$$



And now it's your turn...

A year ago I ran a program on my PC and it took 10 hours.

Now I have three PCs for the same program.

Program initialization, which takes an hour, cannot be parallelized.

What speedup can I expect compared to one PC?

Limitations of Amdahl's Law

Best possible speedup

- Partial parallelism
- Communication overhead
- Task granularity

Summary

- The main components of a computer
- Harvard and von Neumann architectures
- The Fetch-Decode-Execute cycle
- Moore's law
- Amdahl's law

To Do

- Choose a Tutorial Group (Allocation opens at 12, **not** First Come First Serve)
- Familiarize yourself with the Moodle Course
- Read the Assignment Details in the Moodle Course
- Familiarize yourself with Ripes and the Assembly Documentation in the Moodle Course